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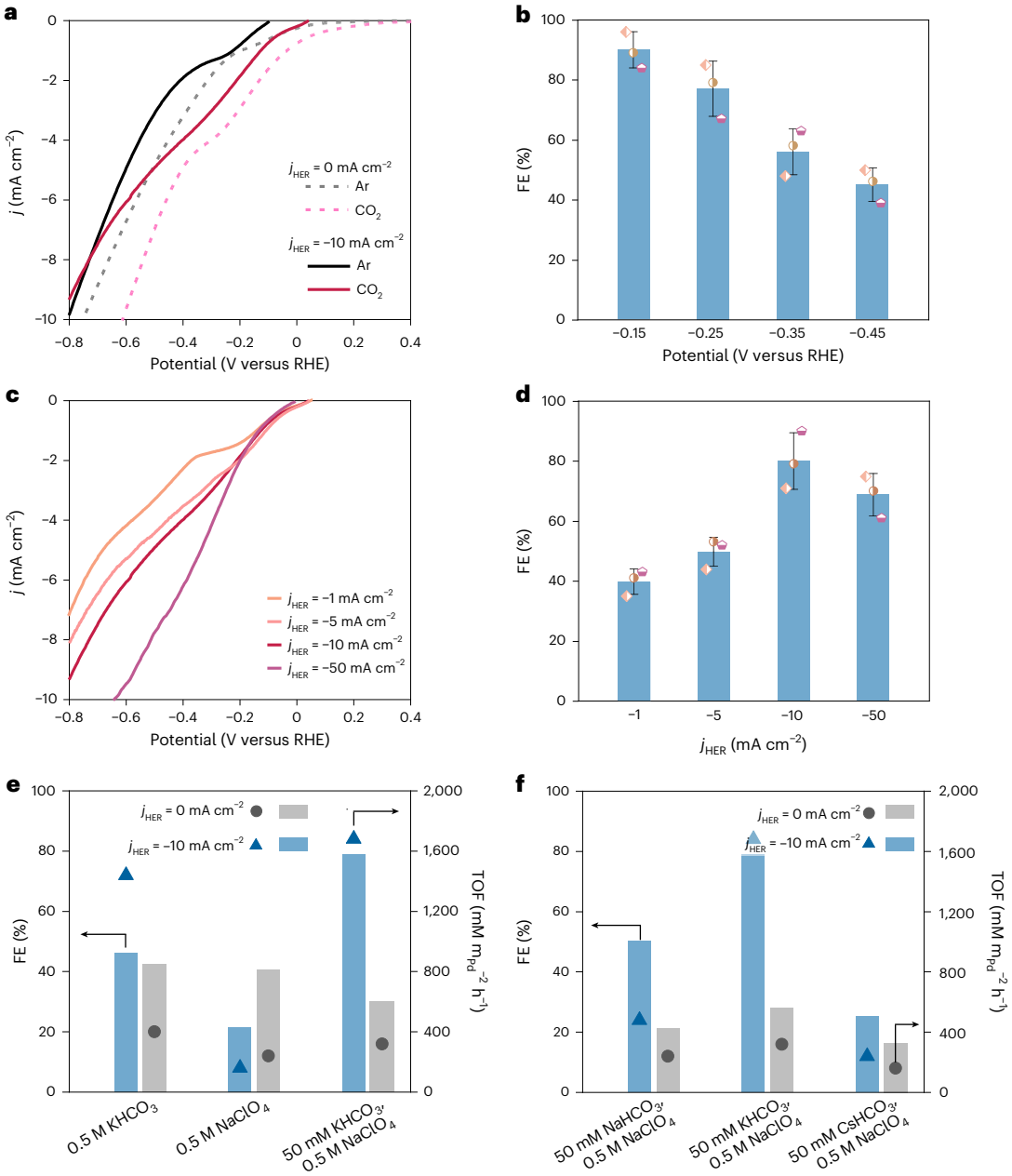


Fig. 2 | Performance of CO_2ER using J-Pd_m under various conditions. a, Linear sweep voltammograms collected on the CO_2ER side of J-Pd_m in the absence or presence of CO_2, together with or without j_{HER} = −10 mA cm^{-2} applied on the HER side. **b**, Faradaic efficiencies of formate production after 3 h of electrolysis at various applied potentials in the CO_2ER chamber with j_{HER} = −10 mA cm^{-2} in the HER chamber. Bars represent mean values, error bars indicate the s. d. and individual data points are overlaid as scatter points (n = 3 independent experiments). **c**, Linear sweep voltammograms collected on the CO_2ER side of J-Pd_m in the presence of CO_2 together with varying j_{HER}. **d**, Faradaic efficiencies of formate production after 3 h of electrolysis at −0.25 V versus RHE in the CO_2ER chamber with varying j_{HER}. Data are presented as mean ± s.d., with individual data points overlaid as scatter points (n = 3 independent experiments). **e**, A performance comparison of CO_2ER after 3 h of electrolysis at −0.25 V versus RHE in the CO_2ER chamber with j_{HER} of 0 versus −10 mA cm^{-2} in various electrolytes of the CO_2ER chamber. **f**, A comparison of the cation effect of CO_2ER after 3 h of electrolysis at −0.25 V versus RHE in the CO_2ER chamber with j_{HER} of 0 versus −10 mA cm^{-2}. In **a**–**f**, the HER chamber electrolyte was 1.0 M KOH in H_2O. In **a**–**d**, the CO_2ER chamber electrolyte was 50 mM $KHCO_3$ and 0.5 M $NaClO_4$ in H_2O.

yielding an electrochemically active surface area approximately 172-fold larger than the original area (Supplementary Figs. 1 and 2 and Supplementary Tables 1 and 2). Such a highly nanostructured surface is beneficial for the efficient accumulation of permeated hydrogen at the Pd–electrolyte–CO_2 triple-phase boundary. In our J-Pd_m design, the CO_2ER chamber is physically isolated from the opposite HER chamber, thus the electrolyte in the HER chamber does not impact the CO_2 reduction in the CO_2ER chamber. Here we decided to utilize 1.0 M KOH in water as the default electrolyte in the HER chamber throughout this study while using 50 mM $KHCO_3$ and 0.5 M $NaClO_4$ in water as the initial electrolyte in the CO_2ER chamber.

As shown in Fig. 2a, it was found that, in the absence of a cathodic current applied on the HER side (that is, without a continuous hydrogen flux), the CO_2ER side of J-Pd_m acts as a conventional Pd electrode. Specifically, in a CO_2-free aqueous electrolyte (that is, 50 mM $KHCO_3$ and 0.5 M $NaClO_4$ in H_2O), the cathodic current was dominated by H_2 evolution.